# AI-Powered Resume Ranker – Project Report

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**Internship Domain: AI And ML Interns**

**Project Duration: 45 Days**

## Introduction

Recruitment processes often involve manually reviewing a large number of resumes, which is time-consuming and inefficient. The AI-Powered Resume Ranker automates this process using Natural Language Processing (NLP) techniques. The system compares resumes with a given job description and ranks candidates according to their relevance and skill match.


## Abstract

The AI-Powered Resume Ranker is a Flask-based web application that uses NLP and Machine Learning techniques to rank resumes automatically. The system extracts text from PDF resumes using PyPDF2 and preprocesses the content using SpaCy by removing stopwords and punctuation and performing lemmatization. TF-IDF Vectorization converts textual data into numerical vectors, while Cosine Similarity measures the similarity between resumes and the job description. The application then displays ranked candidates based on similarity scores, helping recruiters identify suitable candidates efficiently.


## Tools Used

| Tool / Technology | Purpose |
|---|---|
| Python | Core programming language |
| Flask | Web framework for backend and UI |

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|---|---|
| SpaCy | Natural Language Processing |
| Scikit-learn | TF-IDF and Cosine Similarity |
| PyPDF2 | PDF text extraction |
| HTML | Frontend webpage design |
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## Key Insights

- NLP techniques help automate resume screening efficiently.
- TF-IDF converts textual data into numerical vectors for analysis.
- Cosine Similarity measures similarity between resumes and job descriptions accurately.
- Flask provides a lightweight and simple framework for deployment.

## Steps Involved in Building the Project

1. Created a Flask web application for user interaction.
2. Developed a resume upload interface for PDF files.
3. Extracted text from resumes using PyPDF2.
4. Preprocessed text using SpaCy.
5. Applied TF-IDF Vectorization to convert text into vectors.
6. Calculated similarity scores using Cosine Similarity.
7. Ranked resumes and displayed results on the web page.

# Conclusion

The AI-Powered Resume Ranker successfully automates the resume screening process using NLP and Machine Learning techniques. The project reduces manual effort in recruitment and improves candidate selection efficiency. It demonstrates the practical implementation of NLP concepts in real-world applications and can be enhanced further with advanced AI features and improved user interfaces.